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Exercise 1.4.1

Task 1

Suppose you take out a loan with a principal amount of #100,000. If the annual interest rate on the loan is 5% and you plan to repay it in 3 years, how much interest will you owe at the end of the loan term?

For this task write a function called `interest()` that accepts the 3 parameters as input arguments and calculates the amount of interest owed at the end of the loan term. The 3 input arguments should be in the given order:

- The principle loan amount
- The annual interest rate
- The number of years in which the loan should be repayed

Once you have written the function, call it with the provided parameters:

- Principal (P) = #100,000
- Rate (R) = 5% per annum
- Time (T) = 3 years

Solution

Task 1 - Calculating loan interest

we are given:

- principle (P) = 100,000
- Rate (R) = 5% per annum
- Time (T) = 3 years

USINS SIMPLE INTEREST

$\text{interet} = (P \times R \times T)/100$

MANUEL CALCULATION

$\text{interest} = (100000 \times 5 \times 3)/100 = 1500000 / 100 = 15000$

Therefore the interest owed is 15,000.

```
def interest(P,R,T):
    interest_amount = (P * R * T) / 100
    return interest_amount
```

```
### Run your function with the provided parameters here:
P = 100000
R = 5
T = 3

result = interest(P,R,T)
print("The amount owed after", T, "years is", result)
```

The amount owed after 3 years is 15000.0

Task 2

Write a function called `celsius_to_fahrenheit()` that converts a temperature from degrees Celsius to degrees Fahrenheit.

You can use this formula:

$$T_c \times 1.8 + 32$$

where T_c is the temperature in degrees celsius to be converted.

Once you have written the function, use it to convert the following temperatures from Celsius to Fahrenheit:

1. 100
2. 75

3. 120

✓ Solution

Task 2

celsius to fahrenheit

Formula

$$F = (C \times 1.8) + 32$$

$$\text{if } C = 100: F = (100 \times 1.8) + 32 = 212F$$

```
### Run your function with the provided parameters here:
```

```
def celsius_to_fahrenheit(Tc):  
    fahrenheit = (Tc * 1.8) + 32  
    return fahrenheit
```

```
### CG T2
```

```
### Write your function here:
```

```
print ("100 c =",celsius_to_fahrenheit(100),"F")  
print ("75 c =",celsius_to_fahrenheit(75),"F")  
print ("120 c =",celsius_to_fahrenheit(120),"F")  
print ("0 c =",celsius_to_fahrenheit(0),"F")
```

```
100 c = 212.0 F  
75 c = 167.0 F  
120 c = 248.0 F  
0 c = 32.0 F
```

✓ Exercise 1.5.2

✓ Task 1

The number of confirmed COVID-19 cases in selected countries is shown in the table below:

Burundi	694
Ethiopia	113295
Ghana	52198
Kenya	88380
Nigeria	68937
Rwanda	6129
Somalia	4525
South Sudan	3166
Sudan	19196
Uganda	22499

In this task, we are going to practice using and manipulating dictionaries.

1. Create a dictionary for the given data and name it `COVID_19`
2. How many keys are in the data? assign your answer to a variable called `num_keys`.
3. Remove Nigeria from the data
4. Add Tanzania with the value 509

✓ Question 1: Create a dictionary for the given data and name it `COVID_19`

```
### CG T1_1
```

Write your code below:

```
COVID_19 = {  
    "Burundi": 694,  
    "Ethiopia": 113295,  
    "Ghana": 52198,  
    "Kenya": 88380,  
    "Nigeria": 68937,  
    "Rwanda": 6129,  
    "Somalia": 4525,  
    "South Sudan": 3166,  
    "Sudan": 19196,  
    "Uganda": 22499  
}  
  
print(COVID_19)
```

```
{'Burundi': 694, 'Ethiopia': 113295, 'Ghana': 52198, 'Kenya': 88380, 'Nigeria': 68937, 'Rwanda': 6129, 'Somalia': 4525, 'South Sudan': 3166, 'Sudan': 19196, 'Uganda': 22499}
```

- ✓ Question 2: How many keys are in the data? assign your answer to a variable called `num_keys`.

```
### CG T1_2  
### Write your code below:  
num_keys = len(COVID_19)  
  
print(num_keys)
```

```
10
```

- ✓ Question 3: Remove Nigeria from the dataset

```
### CG T1_3  
### Write your code below:
```

```
COVID_19 = {  
    "Burundi": 694,  
    "Ethiopia": 113295,  
    "Ghana": 52198,  
    "Kenya": 88380,  
    "Nigeria": 68937,  
    "Rwanda": 6129,  
    "Somalia": 4525,  
    "South Sudan": 3166,  
    "Sudan": 19196,  
    "Uganda": 22499  
}
```

```
COVID_19.pop("Nigeria")
```

```
print(COVID_19)
```

```
{'Burundi': 694, 'Ethiopia': 113295, 'Ghana': 52198, 'Kenya': 88380, 'Rwanda': 6129, 'Somalia': 4525, 'South Su
```

- ✓ Question 4: Add Tanzania to the dataset. Tanzania recorded 509 cases of COVID-19.

```
### CG T1_4  
### Write your code below:  
COVID_19["Tanzania"] = 509  
  
print(COVID_19)
```

```
{'Burundi': 694, 'Ethiopia': 113295, 'Ghana': 52198, 'Kenya': 88380, 'Rwanda': 6129, 'Somalia': 4525, 'South Su
```

- ✓ Task 2

Consider the dictionary:

```
### CG T2_init
### DON'T DELETE THIS CODE!

food = {
    "fruits": ["apples", "oranges", "pears", "mangoes"],
    "vegetables": ["tomatoes", "lettuce", "spinach", "green peppers"],
    "meat": ["chicken", "fish", "beef", "ostrich"],
    "dairy": ["yogurt", "milk", "cheese", "ice-cream"]
}
```

✓ Question 1: How many keys does the dictionary `food` have?

```
### CG T2_1
### Write your code below:
num_keys = len(food)

print(num_keys)
```

4

✓ Question 2: List the values in `food`

```
### CG T2_2
### Write your code below:
print(food.values())
```

```
dict_values(['apples', 'oranges', 'pears', 'mangoes'], ['tomatoes', 'lettuce', 'spinach', 'green peppers'], ['chicken', 'fish', 'beef', 'ostrich'], ['yogurt', 'milk', 'cheese', 'ice-cream'])
```

✓ Question 3: How do you access "spinach" from food?

```
### CG T2_3
### Write your code below:
print(food["vegetables"][2])
```

spinach

✓ Question 4: How do you add "grapes" to the list of fruits in food?

```
### CG T2_4
### Write your code below:
food["fruits"].append("grapes")

print(food["fruits"])
```

['apples', 'oranges', 'pears', 'mangoes', 'grapes']

✓ Exercise 1.3.1

Task 1

Answer the following using Python:

What is your name? Assign the value to the variable `name`

How old are you? Assign the value to the variable `age`

Are you a male or female? Assign the value to the variable `gender`

What state did you come from? Assign the value to the variable `state`

What language can you speak? Assign the value to the variable `language`

True or False, do you smile today? Assign the value to the variable `smile`

The list of the variables to create and assign values to are:

`name`

`age`

`gender`

`state`

`language`

`smile`

```
### CG T1
### write your code here:

name = "Brain wave"

age = "30"

gender = "male"

state = "Rivers State"

language = "Eleme"

smile = "True"
```

Task 2

Explain why some variable names in the example are acceptable and why some are not in Python

	Sample of acceptable variable names	Unacceptable variable names
1	grade_test	test(grade)
2	covid_19_cases	covid-19-cases
3	term2	2 term
4	Sale_price_2021	2021sales_price
5	miva_learners	miva&learners

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```
### CG T2
```

```
### Write your answer here for sample var
```

Variable naming conventions in Python are strict rules for what the interpreter will accept as a valid "identifier."

```
### Why the "Acceptable" Names Work:
```

All names in the left column followed the rules of Python variables: They start with a letter or an underscore. They only contain alphanumeric characters and underscores (A-z, 0-9, and _). They are not reserved keywords (like if, else, or print).

grade_test: Uses an underscore to separate words, which is perfectly common.

covid_19_cases: Numbers are allowed as long as they aren't the first character.

term2: Ends with a number, which is fine.

Sale_price_2021: Mixes uppercase and underscores legally.

miva_learners: Standard use of letters and underscores.

```
### Why the "Unacceptable" Names Fail
```

Each name in the right column violates a rule of Python variables.

grade(test): Parentheses are used for calling a function, not for variable names. You are trying to run a function named test.

covid-19-cases: (-) is the subtraction operator. You are trying to subtract numbers or variables.

2 term: Variable names cannot start with a number. Allowed use underscores instead, 2 term fails because it can't start with a digit.

2021Sale_price: This violates the rule that

CG T2

Write your answer here for sample variable names

Variable naming conventions in Python are strict rules for what the interpreter will accept as a valid "identifier."

Why the "Acceptable" Names Work:

All names in the left column followed the rules of Python variables: They start with a letter or an underscore. They only contain alphanumeric characters and underscores (A-z, 0-9, and _). They are not reserved keywords (like if, else, or print).

grade_test: Uses an underscore to separate words, which is perfectly common.

covid_19_cases: Numbers are allowed as long as they aren't the first character.

term2: Ends with a number, which is fine.

Sale_price_2021: Mixes uppercase and underscores legally.

miva&learners: The ampersand (&) is a special character. Standard use of letters, numbers, and underscores are all allowed.

Why the "Unacceptable" Names Fail

Each name in the right column violates a specific Python syntax rule:

grade(test): Parentheses are used for calling functions. Python would think you are trying to run a function named test rather than naming a variable.

covid-19-cases: (-) is the subtraction operator in Python. It thinks you are trying to subtract numbers or variables.

2 term: Variable names cannot start with a number. Also, spaces are not allowed. Use underscores instead. 2 term fails because Python variable names can't start with a digit.

2021Sale_price: This violates the rule that a name cannot start with a digit.

miva&learners: The ampersand (&) is a special bitwise operator. Only letters, numbers, and underscores are allowed.

Exercise 1.5.1

13 students have taken an exam. Their scores are: 10, 15, 10, 9, 18, 16, 14, 12, 16, 13, 15, 20, and 17.

You are a teacher and need to help recording these grades.

- Create a list of these scores and assign it to a variable named `marks`.
- Arrange the elements of `marks` in descending order.

✓ Solution

```
# Create a list of scores and name it `marks`.
```

```
# Create a list of the scores
```

```
marks = [10, 15, 10, 9, 18, 16, 14, 12, 16, 13, 15, 20, 17]
```

```
# Arrange the scores in descending order
```

```
marks.sort(reverse=True)
```

```
print("Marks in descending order:", marks)
```

```
# Make sure you print your resulting list
```

```
Marks in descending order: [20, 18, 17, 16, 16, 15, 15, 14, 13, 12, 10, 10, 9]
```

